Graphic Era HILL UNIVERSITY

Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

End Term (Odd) Semester Examination December 2024

Roll no.....

Name of the Course and semester: Electronics & Communication/ 3rd semester
Name of the Paper: Network Theory
Paper Code: TEC -303

Time: 3 hour

Maximum Marks: 100

Note:

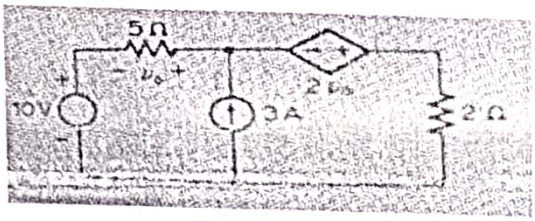
(i) All the questions are compulsory.
(ii) Answer any two sub questions from a, b and c in each main question.
(iii) Total marks for each question is 20 (twenty).
(iv) Each sub-question carries 10 marks.

Q1.

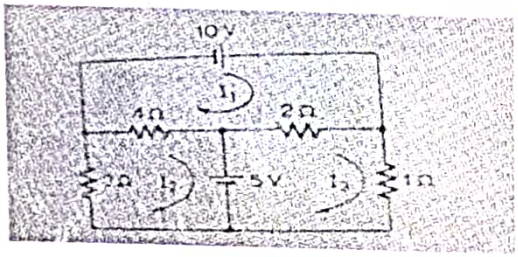
(2X10=20 Marks)

a. State and explain Millmans theorem with suitable example (CO: 1)

b. For the circuit given below find the current in 2 ohm resistor using Nortons theorem. (CO: 1)



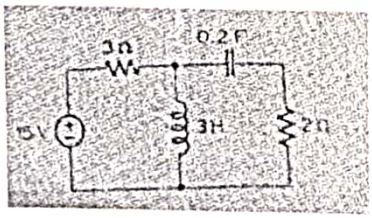
c. Illustrate Tellegens theorem for the network given below. (CO:1)



Q2.

(2X10=20 Marks)

a. Explain the concept of Duality in graph theory also draw the Dual for the Network given below: (CO: 2)



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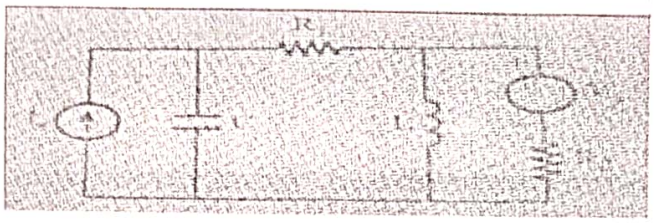
b. For the given reduced incidence Matrix obtain the number of possible trees (CO: 2)

$$[A] = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & -1 & 1 & -1 & 0 & 0 \\ -1 & 0 & -1 & 0 & -1 & 0 \end{bmatrix}$$

c. Draw the graph of the network shown below. How many trees are possible for this graph? Draw any two of your choice. (CO: 2)



Q3. (2X10=20 Marks)

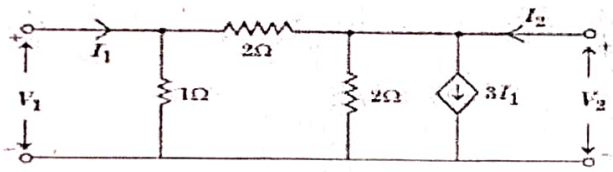
a. Derive the condition of reciprocity and Symmetry for Y parameters. (CO: 4)

b. Perform the following conversions: (CO: 4)

i) Z to Y parameters

ii) Y to H parameters

c. Find Z parameter for the network shown below: (CO: 4)



Q4. (2X10=20 Marks)

a. Realize the following RC driving point impedance function in First Foster form:

$$Z(s) = (s^2+6s+8) / (s^2+4s+3)$$ (CO: 5)

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b. Obtain Develop Cauer-1 form for the function given by: (CO: 5)

$$Z(s) = \frac{s^5 + 5s^3 + 3s}{s^4 + 3s^2 + 1}.$$

c. Write short notes on: (CO: 5)

i) Low pass filter

ii) Band Pass filter

(2X10=20 Marks)

Q5.

a. Define poles and zeros of a transfer function. How do the locations of poles and zeros on the s-plane affect the system's stability and frequency response? (CO: 3)

b. For the given network function, draw the pole zero diagram and hence obtain the time Domain response. (CO: 3)

$V(s) = 5(s+5) / (s+2)(s+7)$

c. Explain The following: (CO: 3)

i) Driving Point and Transfer function

ii) Resonance in AC circuit